

A Multimodal Screening and Validation Framework for Polymer Glass-Transition Temperature Design

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Abstract. The polymer glass-transition temperature (T_g) is a critical parameter governing mechanical and thermal performance as well as the processing window. However, experimental characterization and large-scale molecular dynamics (MD) simulations are often too time- and resource-intensive to support high-throughput screening. Here, we propose a framework for T_g -targeted polymer design that couples generation, multimodal screening, and selective physics-based validation. We adopt pSMILES as a unified structural representation and use wildcard tokens to explicitly denote polymerization sites. For monomer generation, we develop a variational autoencoder that combines a PolyBERT encoder with a Transformer decoder to enable large-scale sampling in a continuous latent space. For T_g prediction, we design a hybrid model that integrates graph neural networks (GNNs) operating on molecular graphs with PolyBERT sequence representations. These complementary modalities are fused via cross-attention, and a prediction-consistency gap (Δ) between unimodal and multimodal predictions is introduced as a lightweight triage criterion within the screening stage before costly downstream validation. On public benchmark datasets, the proposed model outperforms conventional molecular fingerprints and PolyBERT baselines. For further validation, prioritized candidates are evaluated using MD cooling simulations and complementary quantum-chemical calculations. Overall, the key contribution of this work is not a standalone generator or predictor, but a practical inverse-design framework that prioritizes more trustworthy candidates under limited validation resources and provides a transferable basis for future multi-property optimization.

1. Introduction

The glass-transition temperature (T_g) of polymers is a key parameter that links molecular structure, cooperative segmental motion, and macroscopic thermo-mechanical properties.^{1,2} It directly governs the modulus, toughness, and creep behavior of materials at service temperatures, and helps define the processing window of polymer materials.^{3–6} Physically, T_g marks the transition from frozen to thermally activated segmental relaxation.^{2,7} Its value is jointly governed by chain rigidity, intermolecular interactions, and packing/free-volume effects.^{8–12} Therefore, T_g is both a practical design target and a sensitive indicator of how effectively a polymer structure restricts segmental motion.^{13,14}

In recent years, data-driven quantitative structure–property relationship models and machine-learning approaches have emerged as promising tools for polymer T_g prediction and screening. Traditional models based on molecular fingerprints and hand-crafted descriptors are relatively simple and often offer good interpretability, but their ability to capture connectivity-dependent effects and complex structural interactions is limited.^{15,16} Sequence-based representations and polymer language models (*e.g.*, PolyBERT) can better encode chemical context, sequence patterns, and polymerization-related information.^{17,18} However, these representations are still learned mainly from statistical correlations and may not consistently capture polymer-specific structural constraints that are important for T_g , such as repeat-unit regularity and connectivity-dependent packing or interaction patterns.^{16,19} In addition, existing T_g data are often collected under different measurement protocols and conditions, introducing

non-negligible noise and domain shift.^{20,21} Under these conditions, models based on a single representation and point-estimate predictions can become overconfident yet unstable when evaluating out-of-distribution candidates, thereby undermining the reliability of high-throughput screening.^{21,22}

Generative models are increasingly enabling polymer informatics to move from prediction toward design.²³ Early polymer structure generation often relied on sequence-based autoregressive decoding paradigms, typically exemplified by recurrent neural network (RNN) decoders.^{23,24} However, such models often struggle to satisfy both complex syntax and chemical validity simultaneously, often leading to low validity rates, increased duplication, or unstable generation distributions.²⁵ Variational autoencoders (VAEs) provide a unified latent-variable framework for structure generation and latent-space exploration, but the generation quality still depends strongly on the decoder and therefore does not automatically resolve validity constraints.²⁶ More importantly, latent-space sampling in standard VAEs is only weakly coupled to target-property optimization, so generated structures may be valid yet still poorly aligned with a desired T_g .

Based on these challenges, we develop a framework for T_g -targeted polymer design with multimodal screening and physics-based validation. The framework combines unified-representation-based candidate generation, multimodal T_g prediction, consistency-based candidate triage, and MD-based validation within a single pipeline. Specifically, pSMILES is used as a unified structural representation, a VAE is employed for candidate generation in latent space, a multimodal predictor integrates

graph and sequence information for T_g estimation, and a consistency-based candidate-triage criterion is introduced, in which the prediction-consistency gap (Δ) between multimodal and unimodal predictions is used to prioritize generated candidates before costly MD validation. This framework provides a transferable methodological foundation for subsequent multi-objective co-design targeting T_g , mechanical modulus, dielectric constant, and solubility.

2. Methods

2.1. Dataset and Structural Representation

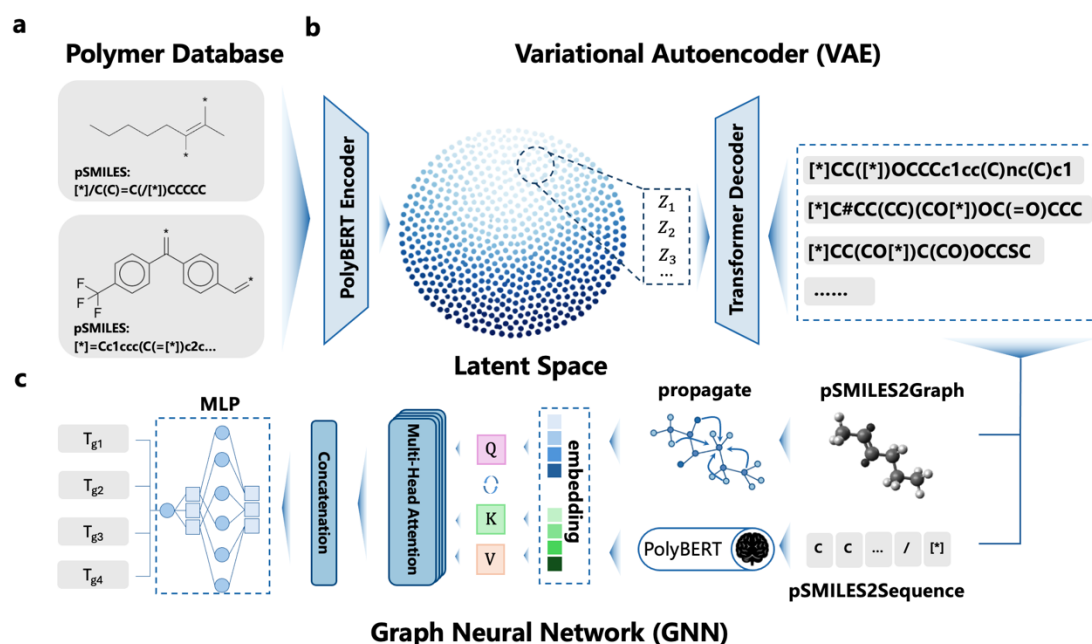


Figure 1. Workflow for T_g -targeted polymer generation and screening. (a) Polymers in the dataset are represented by polymer SMILES (pSMILES), in which wildcard symbols denote polymerization sites. (b) The pSMILES strings are used to train a variational autoencoder (VAE) consisting of a PolyBERT encoder and a Transformer decoder, which learns a continuous latent space for polymer generation. (c) Generated candidates are evaluated using a multimodal T_g prediction model that takes pSMILES-derived graph and sequence representations as inputs. In the graph branch, pSMILES are converted into molecular graphs and encoded by a graph neural

network (GNN). In the sequence branch, pSMILES are processed by PolyBERT. The two representations are fused through a multi-head attention module and passed to a multilayer perceptron (MLP) for T_g prediction.

Two publicly available polymer informatics datasets were used in this study. First, approximately 10^6 polymer structures represented by pSMILES from the PI1M benchmark corpus²⁷ were used to train the variational autoencoder (VAE) in an unsupervised manner. This large corpus enabled the model to learn broad polymer structural distributions. The PI1M dataset, constructed via generative expansion of approximately 12,000 polymer structures collected from PolyInfo²⁸, has been widely used in polymer generation and representation learning.²³ Second, the dataset used for supervised T_g prediction and model evaluation was obtained from the PolyMetriX framework.²¹ The dataset was compiled from T_g records collected from the literature and databases, then standardized and deduplicated. It contains 7,367 polymer samples with standardized pSMILES representations and T_g annotations for supervised property prediction.

To unify the data interface between the generation and prediction modules, we adopted pSMILES as the structural representation of polymers, in which repeat units serve as the core descriptors and wildcard atoms ([*]) explicitly denote polymerization sites. For the generative model, pSMILES were encoded as fixed-length discrete sequences, from which shifted input–output pairs were constructed for autoregressive next-token prediction. For graph construction, pSMILES strings were first parsed using RDKit²⁹, and samples that could not be parsed were excluded from subsequent analyses. Because wildcard atoms cannot directly participate in conventional three-dimensional

conformer generation, the wildcard atoms at polymerization sites were temporarily replaced with carbon atoms while preserving their original bond types to neighboring atoms. The resulting intermediate molecules were then sanitized and hydrogenated. Initial three-dimensional conformers were generated using the ETKDG distance-geometry method³⁰ and further optimized with the MMFF94 force field.³¹ After optimization, the three-dimensional coordinates were mapped back to the atomic indices of the original pSMILES to maintain consistency with the original topological representation.

Based on the parsed structures and generated geometries, each pSMILES was represented as a molecular graph $G = (V, E)$, where nodes correspond to atoms and edges denote chemical bonds. Node features included atomic number, total degree, formal charge, and aromaticity, while three-dimensional coordinates of atoms were retained to characterize local spatial configurations. Edge features included bond type, conjugation, ring membership, and bond length derived from the generated geometry. The resulting graph data were organized together with the corresponding structural identifiers and, when available, Tg labels for subsequent model training and evaluation.

2.2. Polymer Candidate Generation

As illustrated in Fig. 1b, we developed a variational autoencoder (VAE)³² for polymer candidate generation. For the generative model, polymer repeat units were represented as pSMILES sequences, and a pretrained PolyBERT encoder¹⁷ was used to extract sequence-level embeddings. Given an input sequence $\mathbf{x}$, the encoder outputs a

latent representation $\mathbf{h}$, which is further mapped to the mean and log-variance of the latent distribution:

$$\boldsymbol{\mu} = f_{\mu}(\mathbf{h}), \log \boldsymbol{\sigma}^2 = f_{\log \sigma^2}(\mathbf{h}). \quad (1)$$

Sampling was then performed using the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

To retain compatibility with conditional generation, the model included a one-dimensional conditioning interface, in which the scalar condition was mapped by a lightweight multilayer perceptron (MLP) to a conditional latent representation and then fed into the decoder together with $\mathbf{z}$. The decoder was implemented as a multilayer autoregressive Transformer³³, with conditioning information injected through layer-wise feature modulation. During training, teacher forcing was used, and the model was optimized by minimizing the sum of the reconstruction loss and the KL divergence:

$$\mathcal{L} = \mathcal{L}_{rec} + \beta D_{KL}(q(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z})), \quad (3)$$

where $\mathcal{L}_{rec}$ denotes the token-level cross-entropy reconstruction loss and $p(\mathbf{z})$ denotes the standard normal prior.

For large-scale unlabeled training on the PI1M corpus, pSMILES were used as inputs and the conditioning vector set to zero, thereby preserving a consistent model interface while enabling the model to learn the overall polymer structural distribution. To stabilize training, the KL term was linearly annealed to alleviate posterior collapse. AdamW was used together with cosine annealing learning-rate scheduling, gradient clipping, and early stopping based on the validation loss. During inference, new pSMILES candidates were generated by sampling from the latent space and decoding

them autoregressively. Although the generator retained a conditional input interface, conditional generation was not used as the primary strategy in this work. Because a single scalar T_g condition provides only weak guidance in the highly diverse polymer structure space, we adopted a generate-then-screen strategy, in which the generative model was used for broad candidate exploration and the downstream T_g predictor was used for target-range filtering and ranking.

2.3. T_g prediction model

To enable rapid T_g evaluation of both generated candidates and database samples, we developed a multimodal regression model that integrates graph and sequence information. As shown in Fig. 1c, the graph branch takes the molecular graph as input and augments the original bond features with geometry-enhanced edge features derived from interatomic distances and local angular relationships, enabling the joint encoding of local connectivity and three-dimensional structural information. The graph neural network branch was adapted from the gated graph convolution framework proposed by Volgin *et al.*³⁴ Specifically, while retaining the core idea of gated message passing for aggregating neighborhood chemical environments, we developed a hybrid backbone consisting of gated graph convolution layers and edge-aware graph attention layers. After multiple propagation steps, graph-level representations were obtained by combining Set2Set pooling with global mean pooling, which improved encoding of local chemical environments, edge relationships, and the overall polymer structure.

The sequence branch employs PolyBERT to encode pSMILES and extract sequence-level features of polymer repeat units. To jointly model graph and sequence

representations, the two feature streams were first projected into a shared latent space and then aligned through a bidirectional cross-attention module. In this module, sequence features attended to graph features, while graph features attended to sequence features, enabling bidirectional cross-modal information exchange. After masked pooling, the fused cross-modal features were passed to a multilayer perceptron to predict T_g .

T_g prediction was formulated as a supervised regression task. The dataset was divided into training, validation, and test sets using an approximately stratified split based on the T_g distribution. The statistics used for label standardization were computed only from the training set and were used for inverse transformation of the validation and test predictions. During training, mean squared error (MSE) was used as the loss function. AdamW was used together with cosine annealing learning-rate scheduling, gradient clipping, and early stopping based on the validation RMSE. Model performance was primarily evaluated using MAE and RMSE.

2.4. Multimodal Screening and Physics-Based T_g Validation

To support candidate prioritization after chemical-validity filtering, we introduced a multimodal screening step. For each candidate, both the fused-model prediction T_g^{fuse} and the graph-branch prediction T_g^{graph} were obtained, and their absolute difference,

$$\Delta = |T_g^{fuse} - T_g^{graph}|, \quad (4)$$

was used as a coarse-grained consistency signal. Smaller Δ values indicate closer agreement between the two predictions, whereas larger Δ values suggest lower cross-model consistency. Accordingly, Δ was used here not as a direct predictor of MD

agreement, but as a lightweight criterion for deprioritizing lower-confidence candidates before costly downstream validation.

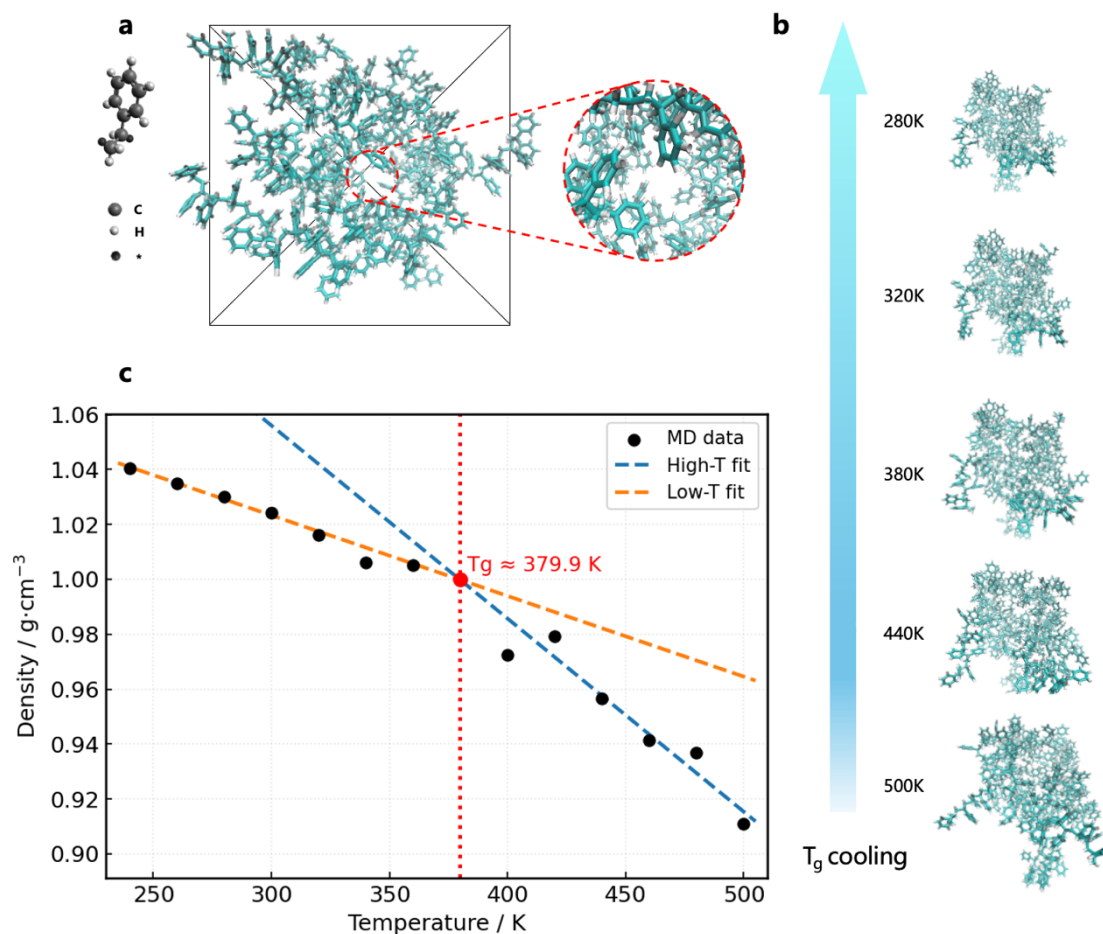


Figure 2. MD-based workflow for T_g estimation, illustrated using polystyrene (PS) as an example. (a) Polymer repeat-unit and equilibrated amorphous simulation cell, with a magnified view of the local chain-packing environment. (b) Representative structural snapshots during stepwise cooling, showing the evolution of chain conformations and packing with decreasing temperature. (c) Density as a function of temperature obtained from the cooling simulation. Linear fits to the high- and low-temperature regimes were used to determine their intersection, which was taken as the MD-estimated T_g . Black circles denote the MD data, blue and orange dashed lines denote the two linear fits, and the red marker with the vertical dotted line indicates the extracted T_g for the illustrative PS case.

For prioritized candidates, molecular dynamics (MD) simulations were performed for physics-based validation. Polymer structures were first generated in RadonPy³⁵ from pSMILES descriptors. Atomic partial charges were then derived by RESP fitting to electrostatic potentials calculated with Psi4³⁶ at the HF/6-31G* level, and force-field parameters were assigned using GAFF2_mod. Amorphous cells were subsequently constructed and equilibrated using the preset EQ21step protocol, with additional NPT relaxation applied when necessary to obtain stable initial configurations. All equilibration and stepwise NPT cooling simulations were conducted in LAMMPS.³⁷ The glass-transition temperature (T_g) was extracted from the simulated density–temperature relationship by bilinear fitting of the high-temperature rubbery regime and the low-temperature glassy regime. The intersection of the two fitted lines was defined as the MD-estimated T_g .

To further probe the structure–property relationships of representative candidates, quantum-chemical calculations and surface analyses were performed on the corresponding monomer molecules. Quantum-chemical calculations were performed on the HZWCloud platform using the BDF program³⁸, and the resulting wavefunctions were analyzed with Multiwfn.³⁹ In parallel, VMD⁴⁰ was used to visualize the spatial distributions of molecular orbitals and electrostatic potential (ESP), allowing examination of the polarity distributions, frontier-orbital characteristics, and local interaction tendencies from an electronic-structure perspective. These quantum-chemical calculations were used primarily for mechanistic interpretation rather than direct numerical validation of T_g . They were intended to help rationalize how

functional-group environments, electronic delocalization, and surface electrostatic potential distributions may influence constraints on segmental motion and the observed T_g trends.

3. Results

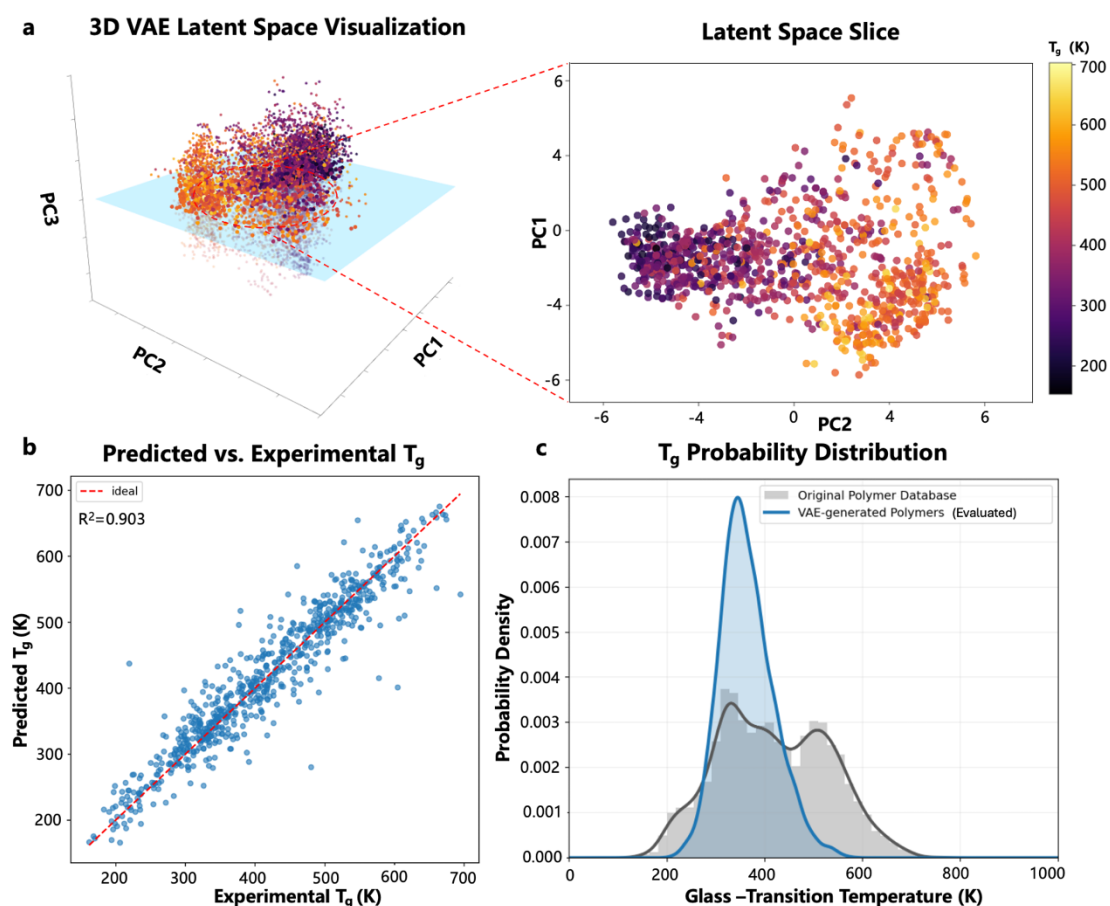


Figure 3. VAE latent-space, T_g prediction performance, and T_g distribution of generated polymers. (a) Latent representations learned by the VAE, visualized in principal-component space. A three-dimensional projection onto PC1-PC3 is shown on the left, together with a two-dimensional latent-space slice in the PC1-PC2 plane on the right. Point colors denote the corresponding T_g values. **(b)** Comparison between predicted and experimental T_g values on the test set, with the red dashed line indicating ideal agreement. **(c)** Comparison of the T_g probability distributions of the original polymer database and VAE-generated polymers evaluated by the multimodal predictor.

The PolyBERT-Transformer VAE trained on the PI1M corpus was able to stably generate chemically valid pSMILES candidates under large-scale unconditional sampling. After RDKit parsing and rule-based chemical filtering, the generated samples maintained a validity of approximately 0.90. In a representative sampling run, the uniqueness reached about 0.998 and the novelty approached 1.0, indicating that nearly all generated structures were absent from the training set. These results show that, despite the strict grammatical constraints of polymer pSMILES, the generator achieves a favorable balance of validity, uniqueness, and novelty, providing a sufficiently large candidate pool for downstream high-throughput screening.

As shown in Fig. 3a, the VAE latent representations were visualized in principal-component space by a three-dimensional projection together with a two-dimensional latent-space slice. These representations do not form sharply separated clusters but instead define a relatively continuous manifold. When the latent representations are colored by predicted T_g , a discernible property gradient emerges along the latent coordinates, suggesting that the learned manifold captures structure-dependent trends relevant to T_g . These observations suggest that, even without explicit conditional generation, the VAE latent space learned can still provide meaningful structural priors for subsequent property-guided candidate screening.

Furthermore, the predicted T_g distribution of the generated candidates partially overlapped with that of the original polymer database, although the two distributions were not identical. As shown in Fig. 3c, compared with the original database, the VAE-generated samples evaluated by the predictor were more concentrated in the

intermediate T_g region, exhibited a narrower overall distribution, and showed limited coverage of the high- T_g tail. This result suggests that the generated samples retain partial overlap with the property distribution of known polymers and do not deviate drastically from the original data domain. However, coverage of the target T_g space, especially in the high- T_g region, remains limited, indicating that subsequent target-oriented screening and physics-based validation are still needed to focus on more valuable candidates.

Table 1. Comparison of T_g prediction performance of different models on the PolyMetriX test set. The models compared include conventional machine-learning models, PolyBERT-based sequence models, graph neural network models, and the graph–sequence fusion model, all evaluated under the same data split. The evaluation metrics include root-mean-square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2); lower RMSE and MAE and higher R^2 indicate better predictive performance.

Artificial Intelligence (AI) Model	RMSE	MAE	R^2
Random Forest	49.43	37.06	0.808
Support Vector Regression (SVR)	57.75	45.03	0.737
Multi-Layer Perceptron (MLP)	52.24	39.93	0.785
PolyBERT SVR	48.71	35.73	0.813
PolyBERT MLP	40.69	27.58	0.870
PolyGeoGAT	37.46	25.57	0.890
CrossAttentionPolyGeoGAT	35.12	24.27	0.903

On the PolyMetriX test set, the proposed T_g prediction model showed strong regression performance. As shown in Fig. 3b, the predicted and experimental values

were generally distributed along the ideal diagonal, indicating that the model captured the structure- T_g relationship reasonably well. However, the scatter became more dispersed in the higher- T_g region, suggesting increased prediction uncertainty caused by data sparsity or insufficient representation in that regime.²¹ The quantitative comparison is summarized in Table 1. Conventional machine-learning baselines, including Random Forest, SVR, and MLP, exhibited relatively large prediction errors overall. Incorporating PolyBERT representations led to a clear improvement in prediction performance, with PolyBERT-MLP achieving an MAE of 27.58 K, an RMSE of 40.69 K, and an R^2 of 0.870. The graph-based PolyGeoGAT further improved the prediction accuracy, achieving an MAE = 25.57 K, an RMSE = 37.46 K, and an R^2 = 0.890. CrossAttentionPolyGeoGAT, which fuses graph representations and sequence features through cross-attention, achieved the best overall performance, with an MAE = 24.27 K, an RMSE = 35.12 K, and an R^2 = 0.903.

The comparative results suggest two main conclusions. First, explicit graph-based message passing better captures the local chemical environments, backbone topology, and associated geometric information relevant to T_g . Second, sequence-based and graph-based representations are complementary, and cross-modal fusion can further improve predictive accuracy and prediction stability. Therefore, the fused model can serve as an effective scoring function for the subsequent generation-screening framework.^{41,42}

Table 2. Comparison between multimodal predicted T_g values and MD-estimated T_g values for representative generated polymer candidates selected after screening.

pSMILES	Model T_g	MD T_g	Deviation %
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[*]NNC(=O)c1ccc(Cc2cc(S(=O)(=O)Nc3ccc([*])cc3)ccc2C)c(N=C)c1	492.2	454.3	8.36
[*]Oc1ccc(NC(=O)c2ccc(CC(=O)Oc3ccc([*])cc3)cc2)c2ccccc12	476.7	444.3	7.30
[*]OS(=O)(=O)c1ccc(Cc2ccc(N3C=Cc4ccc([*])cc4C3=O)cc2)c(-c2ccccc2)c1	459.8	469.6	2.07
[*]Oc1ccc(C(=O)c2ccc(Nc3cccc(-c4ccc(S([*])(=O)=O)cc4)c3)cc2)cc1	456.4	463.8	1.58
[*]c1ccc(CCCCCOc2ccc(-n3c([*])c4cccc4c3)cc2S(=O)(=O)O)c(Cl)c1	443.9	415.5	6.84
[*]CC(O)COc1ccc(C(C)(C)c2ccc(OC(=O)c3cc(OC(=O)cc(C([*])=O)c3)cc2)c(-c2ccccc2)c1	399.2	430.1	7.16
[*]/C=C/c1cccc1CCC(O)CNC(=O)c1ccc(N(C(=O)CCCCCOc2ccc([*])c(O)c2)cc1)c1ccccc1	389.3	413.2	5.76
[*]COc1ccc(N(CC)c2ccc(C(=O)c3cccc(OC(=O)SCCC([*])(C)C)c3)cc2)cc1	387.4	411.1	5.76
[*]OP(=O)(O)OCCOC(=O)c1cccc(C(C)(C)c2ccc(OC(=O)c3ccc([*])cc3)cc2)c1CCC	338.9	455.6	25.6
[*]CC([*])(C)COc1ccc(C(=O)OCCCCC(CO)=Nc2ccc(OC(=O)c3ccccc3)cc2)cc1	336.9	408.8	17.58
[*]CC([*])C(=O)OCC(C)(C)CCCCCOc1ccc(S(=O)(=O)c2ccc(OC)cc2)c1	303.7	360.1	15.66
[*]CCN(CCCC([*])(C#N)CSC(=O)OCCCC)NC	302.4	321.8	6.02
[*]CC([*])OC1CC1N(C)c1ccc(C(=O)Oc2ccc(S(=O)(=O)CCC)cc2)s1	350.5	440.9	20.49
[*]C/C=C/c1ccc(C(CBr)(COC(=O)c2ccc(Cl)cc2)OCCCCCCC)c([*])c1	340.5	385.9	11.75
[*]c1ccc(OCCOc2ccc(C(C)(C)c3ccc([*])cc3)cc2)cc1	368.3	365.3	0.82
[*]CC([*])c1ccc(C(=O)NCCC)cc1/C=N/c1ccc(O)cc1	398.2	455.2	12.51
[*]/C(=N/c1ccc(C(=O)S[*])c(OCCCC(F)(F)F)c1)ccccc1	395.9	442.6	10.53
[*]Cc1ccc(/C=C/c2ccc(OC(=O)c3cc(CCl)cc([*])c3)cc2OCCCC)cc1	409.7	433.1	5.38
[*]NC(CO)C(=O)Nc1cccc(CC([*])=O)c1	448.4	485.8	7.69
[*]CC(=O)Oc1ccc(S(=O)(=O)c2ccc(OC(=O)c3ccc(C)cc3)cc2)c(OC([*])=O)c1	443.9	484.7	8.39
[*]Cc1ccc(CS(=O)(=O)c2cc(C([*])(C)C)cc(-c3ccccc3)c2)cc1C	475.9	489.2	2.70

Table 2 summarizes 21 polymer candidates selected from the VAE-generated set, together with their pSMILES representations, multimodal predictions, MD-estimated T_g , and the relative deviation. To examine whether prediction consistency carries practical information for generated-space screening, we compared the prediction-consistency gap (Δ) with the deviation between multimodal predictions and MD-estimated T_g for representative candidates. Although the relationship was not strictly monotonic at the individual-sample level, smaller Δ values generally enriched the

screened set with candidates showing lower MD deviations on average. This suggests that Δ is best interpreted as a coarse-grained triage criterion rather than a standalone predictor of MD agreement. The relationship between Δ and MD deviation is provided in Table S1.

To assess the applicability of the predictor in the generated chemical space, representative prioritized candidates were selected for MD-based validation, and the **predicted T_g values** were compared with the MD-estimated T_g values. Fig. 2 uses polystyrene (PS) as an illustrative example of the MD cooling protocol and the bilinear fitting procedure used for T_g extraction. Applying the same protocol to the selected generated candidates yielded clear density-temperature transitions, from which candidate-specific MD-estimated T_g values were obtained. The comparison results for 21 representative candidates are summarized in Table 2. Overall, the MD-estimated T_g values and model predictions showed generally consistent trends, although a systematic deviation remained. Relative to the MD results, the predicted values were overall lower, with a mean error of approximately -29.14 K, an MAE of 38.84 K, and an RMSE of 47.46 K, together with a few outliers exhibiting relatively large deviations.

These results show that the machine-learning model provides a practical basis for preliminary screening and ranking in a large candidate space, although its errors on generated candidates, especially potentially out-of-distribution samples, remain non-negligible. Accordingly, predictive consistency and target matching were combined for candidate prioritization, while MD was used as a downstream physics-based validation tool to confirm the most promising structures under a limited computational budget.

Meanwhile, the systematic deviations revealed by MD also provide a basis for analyzing potential error sources, such as insufficient data coverage in the high- T_g region, limited extrapolation capability of the model, and systematic shifts caused by MD cooling rates and finite-size effects.

4. Discussion

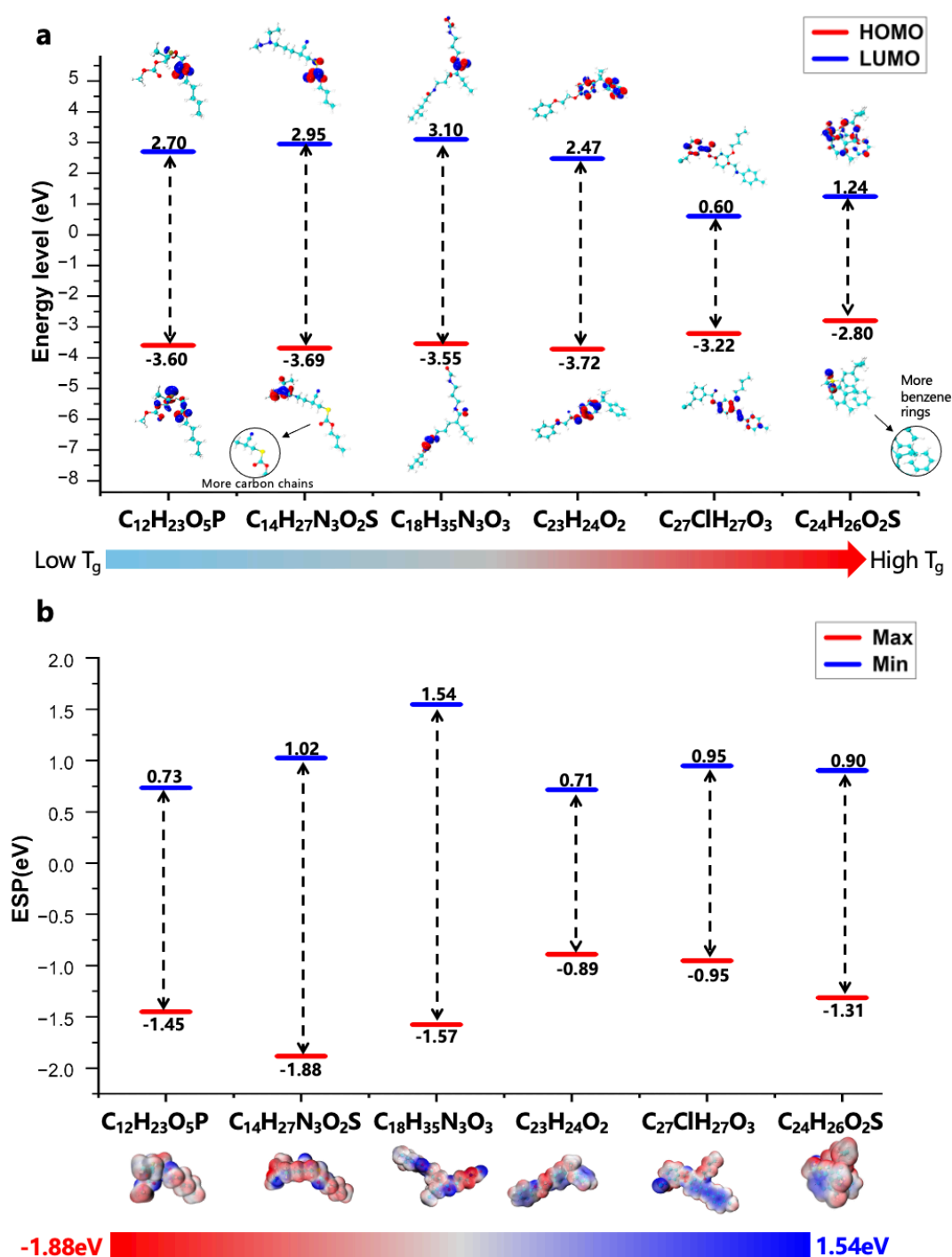


Figure 4. Frontier-orbital energies and electrostatic-potential characteristics of representative polymer candidates spanning the T_g range. (a) HOMO/LUMO energy levels and corresponding frontier-orbital isosurfaces for six representative candidates arranged from low T_g to high T_g . The dashed arrows denote the HOMO–LUMO gaps, and selected annotations highlight characteristic structural motifs. **(b)** Surface electrostatic potential (ESP) analysis for the same set of candidates. The upper and lower values denote the positive and negative ESP extrema, respectively, and the mapped surfaces visualize the spatial distribution of electrostatic potential from negative (red) to positive (blue).

This work establishes a computational screening framework for T_g -targeted polymer design, in which low-cost generation and multimodal screening are used for large-scale exploration, whereas MD simulations are reserved for small-scale downstream validation.^{41–43} This division of labor improves screening efficiency while reducing reliance on a single point-estimate predictor, making the framework better suited to practical polymer discovery under limited validation resources.

Although the generator retained a conditional input interface, direct conditional generation was not used as the primary mechanism for target-directed design. Because T_g arises from multiple coupled factors^{1,2,8–12} and corresponds to a highly one-to-many mapping in polymer structure space, a single scalar condition is insufficient to impose reliable structural constraints. Under the current data scale and label quality, a generation-screening strategy is therefore more robust. Under this strategy, target-directedness is introduced progressively through prediction, consistency-based candidate triage, and physics-based validation rather than being enforced during generation itself. In particular, the prediction-consistency gap (Δ) is used here as a

practical consistency-based signal for candidate triage, helping deprioritize potentially lower-confidence samples before costly MD or experimental validation.⁴⁴ Thus, Δ is not treated as a new property target, but as a practical tool for risk-aware candidate prioritization under limited validation resources.⁴⁵

MD-based validation further clarified the scope and limitations of the present framework. The model remains useful for relative ranking in the generated space, but its numerical accuracy is still limited for potentially out-of-distribution candidates. The observed deviations likely arise from sparse training coverage in the high- T_g region, the fact that the predictor is trained on database-reported T_g labels, and systematic offsets in MD estimates associated with simulation protocol and cell construction. MD should therefore be viewed here as a post hoc physical reference rather than as the sole ground truth. Within this context, complementary HOMO/LUMO and ESP analyses provide a consistent mechanistic picture of the differences among the representative candidates. Higher- T_g candidates tend to exhibit smaller HOMO/LUMO gaps together with milder and more spatially uniform surface electrostatic potentials, whereas lower- T_g candidates more often show larger gaps and stronger local charge separation. These features suggest that higher T_g is associated not with any single electronic descriptor, but with a broader combination of stronger conjugation, greater delocalization, more rigid backbones, and more continuous weak intermolecular interactions, all of which may help strengthen constraints on segmental motion and interchain packing.^{46,47} These quantities should therefore be interpreted as correlated electronic signatures of

the underlying structure–dynamics relationship, rather than as direct determinants of T_g .

5. Conclusion

In summary, this work establishes a framework for T_g -targeted polymer design that integrates candidate generation, multimodal screening, and physics-based validation. The framework balances exploration efficiency with trustworthy candidate prioritization, and thus provides a practical and extensible basis for data-driven polymer inverse design.

MD-based validation showed that the machine-learning predictions and MD-estimated T_g values were generally consistent in trend, although systematic deviation remained. This result highlights the practical value of the generation–prediction–physics-based validation strategy for candidate prioritization and bias identification. Combined with the quantum-chemical analysis, these results further suggest that higher- T_g candidates tend to be associated with greater backbone rigidity, more restricted conformational freedom, stronger electronic delocalization, and more uniform surface electrostatic potential distributions. Overall, this work provides a computational framework for the high-throughput inverse design of T_g -targeted polymers that balances efficiency, trustworthy candidate prioritization, and physical interpretability, while offering a methodological basis for future multi-property polymer design targeting modulus, dielectric constant, solubility, and related properties.

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Author contributions

M.Y. conceived and supervised the project. T.H. and W.B. implemented the VAE generation model and the multimodal T_g prediction model, performed model training and evaluation. W.B. conducted molecular dynamics. T.H. conducted quantum-chemical analyses, and prepared the original draft and visualizations. M.Y. contributed to result interpretation and manuscript revision. All authors participated in discussion of the results and approved the final manuscript.

Code and Data Availability

The code used in this study is available through the GitHub repository *PolyMolStudio* at <https://github.com/geinitangwanle/PolyMolStudio>. The repository contains the implementation of the pSMILES-based polymer generation framework, the T_g prediction model, and the associated screening framework.

Raw VAE-Generated Polymer-Monomer Candidate Library is available via figshare at <https://doi.org/10.6084/m9.figshare.31821124>. Supplementary Data S1 is provided as a separate CSV file containing the full raw candidate library generated by the VAE model in this study. The dataset includes 2,000,000 generated polymer-monomer candidates represented as SMILES strings with polymerization-site markers ([*]). Among them, 1,800,738 were chemically valid, 1,995,121 were unique, and 1,994,891 were novel relative to the training set, corresponding to validity, uniqueness, and novelty values of 90.03%, 99.76%, and 99.99%, respectively. For each entry, the file records a unique sample identifier (sample_id), the generated structure (smiles), and three Boolean annotation fields indicating chemical validity (is_valid), uniqueness within the generated set (is_unique), and whether the structure is already present in the training set (in_training_set). This dataset constitutes the initial candidate pool for subsequent filtering, screening, and downstream validation.

Competing interests

The authors declare no competing interests.

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